A Comprehensive Review of Proximal Electromagnetic Sensors' Accuracy and Cost Considerations for Soil Property Prediction and Mapping

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VI Global Proximal Soil Sensing Workshop

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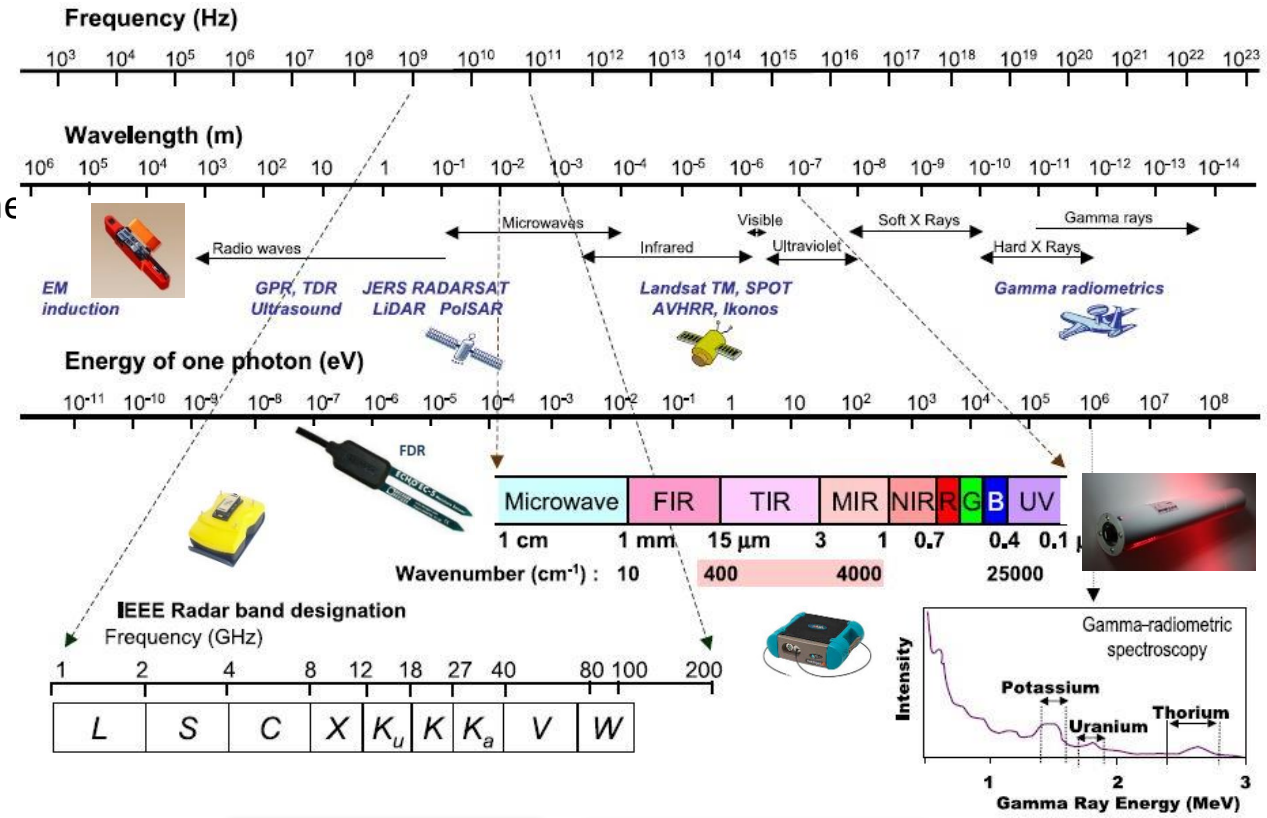
INTRODUCTION: PROXIMAL SOIL SENSORS

Electromagnetic PSSs provide quantitative results that can be more **time- and cost-effective** than conventional laboratory analyses (Viscarra Rossel et al., 2011)

Depending on the section of the electromagnetic spectrum the are sensitive to,

- γ-radiometrics (**γ-ray**)
- X-ray fluorescence (**XRF**)
- Vis-NIR-MIR spectroscopy (**DRS**)
- Ground-penetrating Radar (**GPR**)
- Electromagnetic Induction (**EMI**)
- Time and Frequency-domain Reflectometry (**TDR/FDR**)

- **Laboratory or portable** sensors
  - Point measurements
  - Rapid estimations
- **On-the-go** sensors
  - Spatial variation of properties
  - Topsoil and subsoil maps





Portable VisNIR
(ASD Fieldspec)



On-the-go
(DUALEM 21H)

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BENEFITS OF USING PROXIMAL SOIL SENSORS

Spatialization of properties
Soil fertility
Soil health

- SOC
- Nutrients
- Soil elemental composition
- Mineralogy
- Electrical conductivity
- Water content
- Soil texture
- Soil depth
- Cemented horizons
- Soil bulk density
- Exchangeable bases
- CEC
- Acidity

Precision agriculture (users and stakeholders)

- High-resolution soil maps
- Reduction of costs compared to conventional analysis and labor-intensive sampling
- Making informed decisions
- Improvement of crops productivity
- Optimization of resources for crop necessities

Policy-makers

- Monitoring
- Reporting
- Verification
- Soil health conservation
- Mitigation of impacts

Academics

- Modeling techniques
- Future projections
- Research and development

The ProbeField project: establish protocols for robust in-field monitoring of carbon stock and soil fertility based on proximal soil sensors (PSS)

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JUSTIFICATION: QUESTIONS TO ANSWER

GAPS / Questions	PA (users+stakeholders)	Policy-makers	Academics
What accuracy can we expect from PSS?	X	X	X
How accurate are these surveys compared to conventional soil	X	X	X
P<0.05????			X
Are there ways to improve the accuracy of PSS? (use of covariates)		X	X
How much does all this cost?	X	X	X

Aim of the review: to provide insights relevant to choosing the most suitable PSS technique through a comprehensive review of available literature on costs and accuracies of key soil property measurements

WORKFLOW

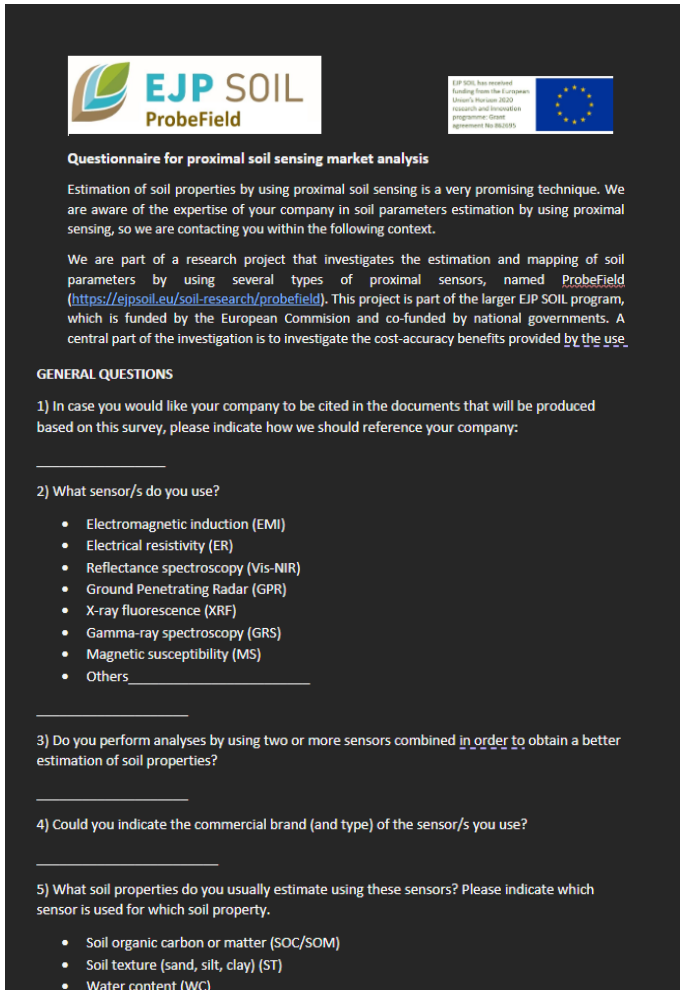
Search engines:
E.g., GS, RG, Scopus

Keywords:
E.g., PSS, estimation, soil map, cost, covariates,

Selection criteria:
E.g., accuracy measurements

Lack of information on costs

Questionnaire directed to PSS companies



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Questionnaire for proximal soil sensing market analysis

Estimation of soil properties by using proximal soil sensing is a very promising technique. We are aware of the expertise of your company in soil parameters estimation by using proximal sensing, so we are contacting you within the following context.

We are part of a research project that investigates the estimation and mapping of soil parameters by using several types of proximal sensors, named ProbeField (https://ejpsoil.eu/soil-research/probefield). This project is part of the larger EJP SOIL program, which is funded by the European Commision and co-funded by national governments. A central part of the investigation is to investigate the cost-accuracy benefits provided by the use.

GENERAL QUESTIONS

1) In case you would like your company to be cited in the documents that will be produced based on this survey, please indicate how we should reference your company:

_____

2) What sensor/s do you use?

- Electromagnetic induction (EMI)
- Electrical resistivity (ER)
- Reflectance spectroscopy (Vis-NIR)
- Ground Penetrating Radar (GPR)
- X-ray fluorescence (XRF)
- Gamma-ray spectroscopy (GRS)
- Magnetic susceptibility (MS)
- Others_____

_____

3) Do you perform analyses by using two or more sensors combined in order to obtain a better estimation of soil properties?

_____

4) Could you indicate the commercial brand (and type) of the sensor/s you use?

_____

5) What soil properties do you usually estimate using these sensors? Please indicate which sensor is used for which soil property.

- Soil organic carbon or matter (SOC/SOM)
- Soil texture (sand, silt, clay) (ST)
- Water content (WC)

Accuracy

- 115/209 papers
- **NRMSE = RMSE/y_{max}-y_{mim}**
  - For PSS individually
  - In combination
  - Effect of covariates
  - Effect of modeling techniques (statistics)
  - Accuracy of 3D-mapping (subsoil)

Costs

- 24 of companies in 11 countries
- **Questions**
  - What sensors do they use?
  - What are the most important properties?
  - What properties are most difficult to measure?
  - Average cost per ha and per working day?
  - Cost distribution?

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GROUPING PROCESS OF PROPERTIES

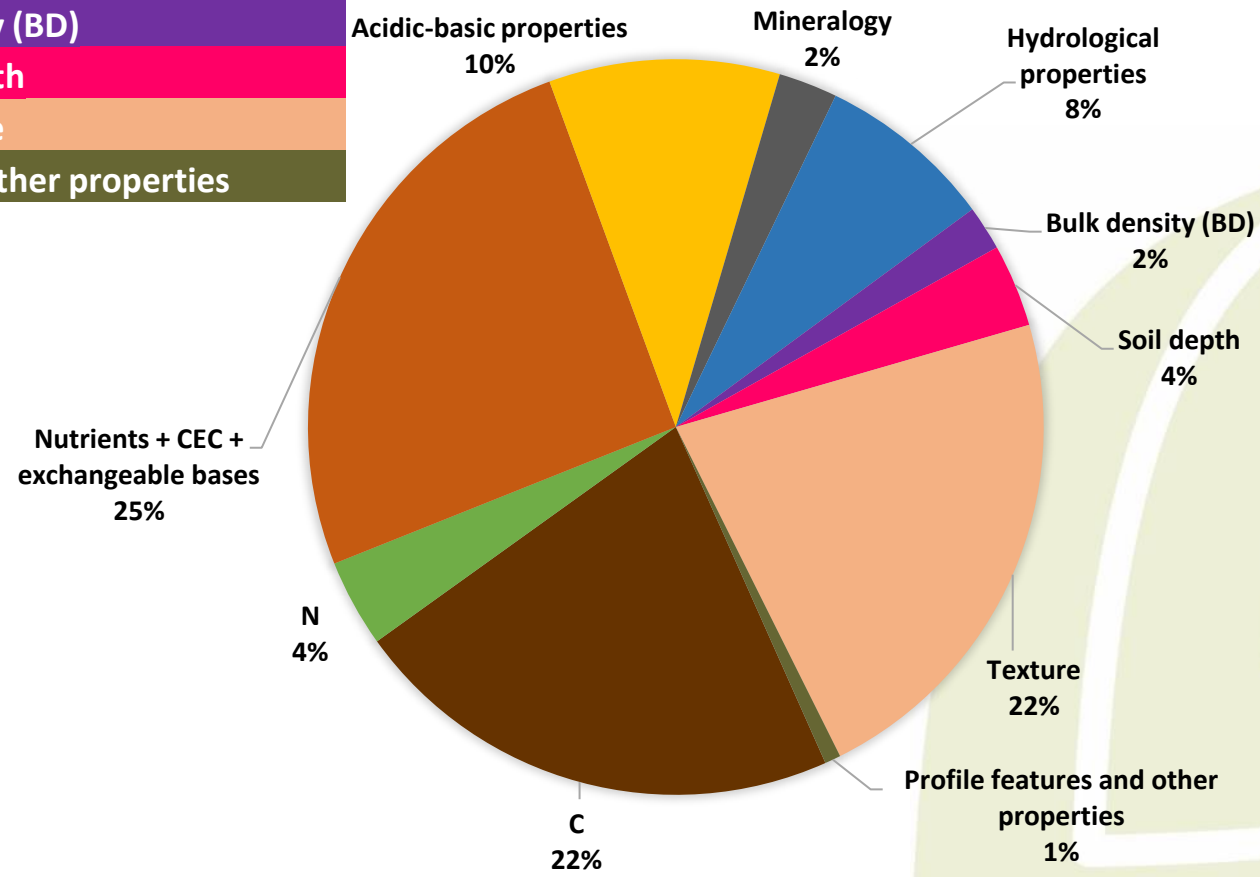
Soil properties from references

- Total carbon (TC)
- Soil organic matter (SOM)
- Soil organic carbon (SOC)
- Total Nitrogen (N)
- Phosphorous (P)
- Potassium (K)
- Cation exchange capacity (CEC)
- Plant available potassium (paK)
- Exchangeable Magnesium (ex-Mg)
- Exchangeable Calcium (ex-Ca)
- Exchangeable Phosphorous (ex-P)
- Exchangeable Potassium (ex-K)
- Other secondary nutrients
- Soil inorganic carbon (SIC)
- Carbonates (CaCO3)
- Salinity (ECe)
- Exchangeable Aluminium (ex-Al)
- Base saturation percentage (BSP)
- Electrical conductivity (EC)
- Lime buffer capacity (LBC)
- pH
- H+Al
- Other chemicals (e.g., heavy metals)
- Clay type
- Moisture
- Volumetric water content (VWC)
- Field capacity
- Evapotranspiration
- Wilting point
- Bulk density (BD)
- Soil depth
- Coarse fragments
- Silt (texture)
- Sand (texture)
- Clay (texture)
- Horizons
- Compacted layer
- Soil structure
- Penetration resistance
- Total porosity
- Biological properties

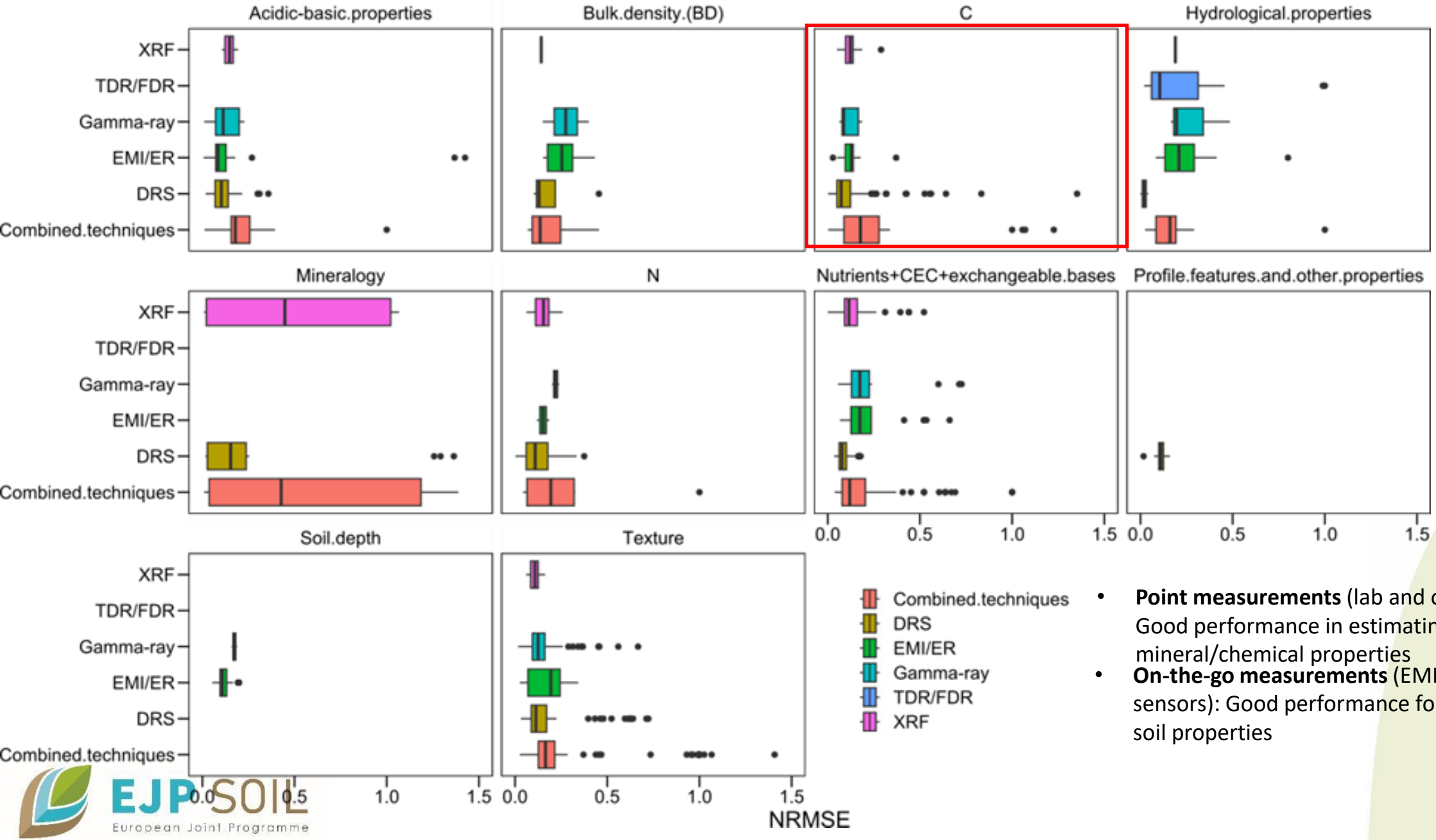
Groups of soil properties

- C
- N
- Nutrients + CEC + exchangeable bases
- Acidic-basic properties
- Mineralogy
- Hydrological properties
- Bulk density (BD)
- Soil depth
- Texture
- Profile features and other properties

- >1500 records
- PSS techniques
- Covariates
- Topsoil/subsoil estimates
- Modeling technique



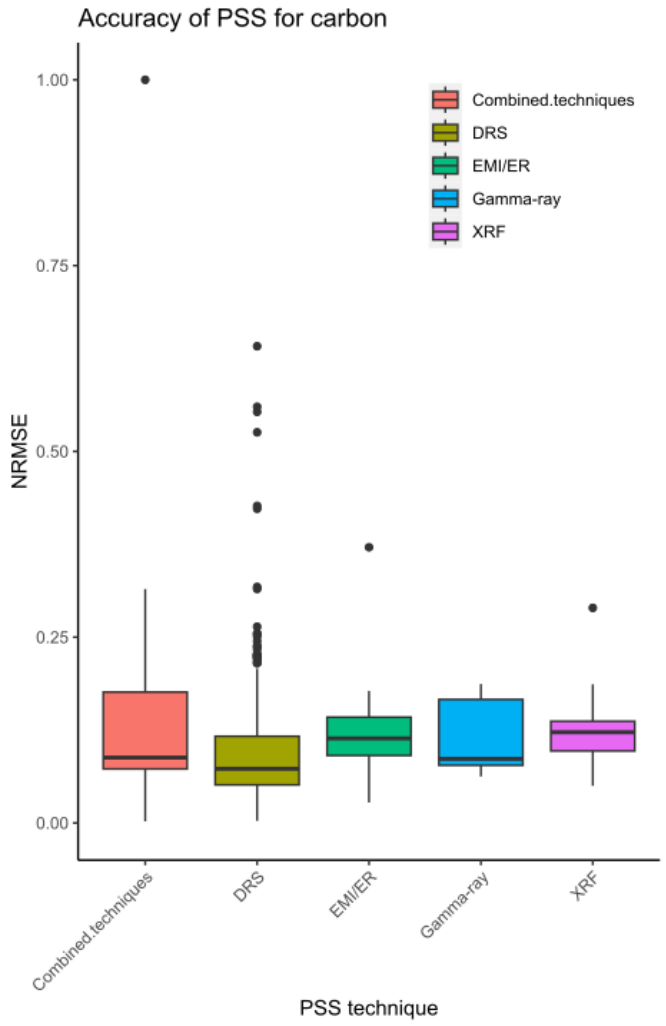
ACCURACY OF PSS TECHNIQUES



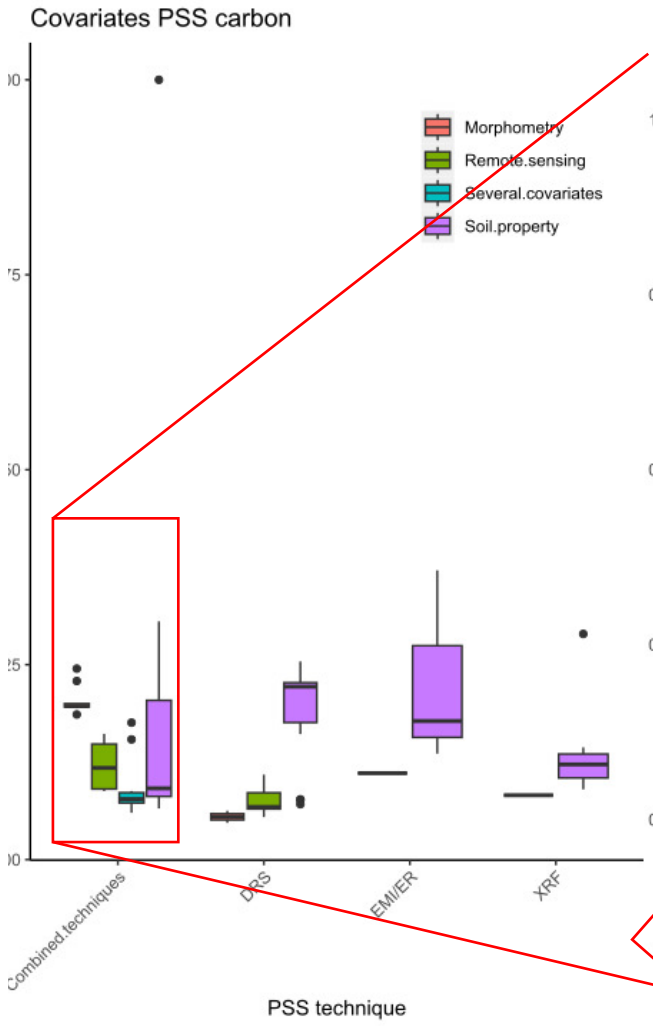
- **Point measurements** (lab and on-the-field): Good performance in estimating mineral/chemical properties
- **On-the-go measurements** (EMI and gamma-ray sensors): Good performance for spatialization of soil properties

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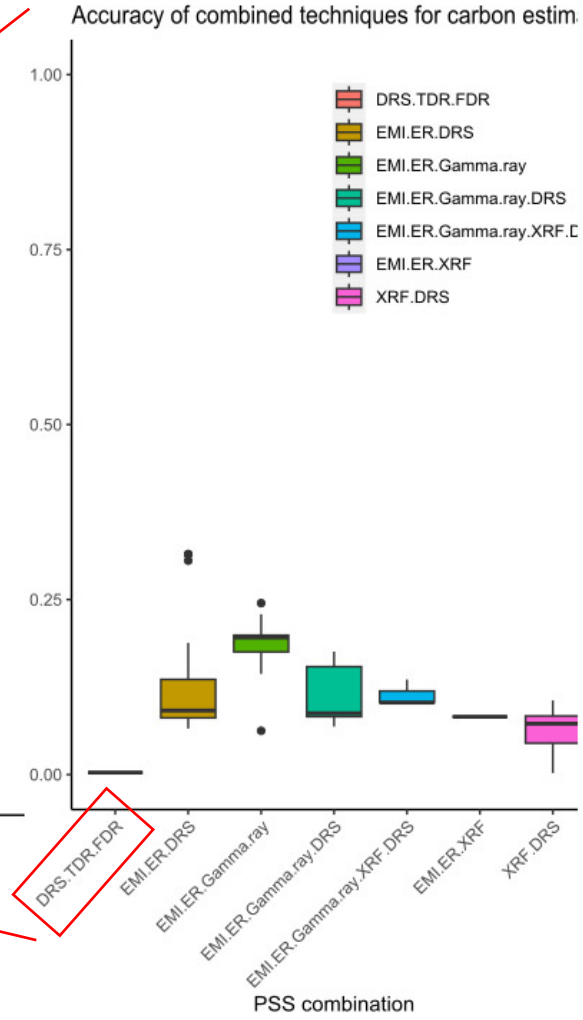
ESTIMATING SOC WITH PSS



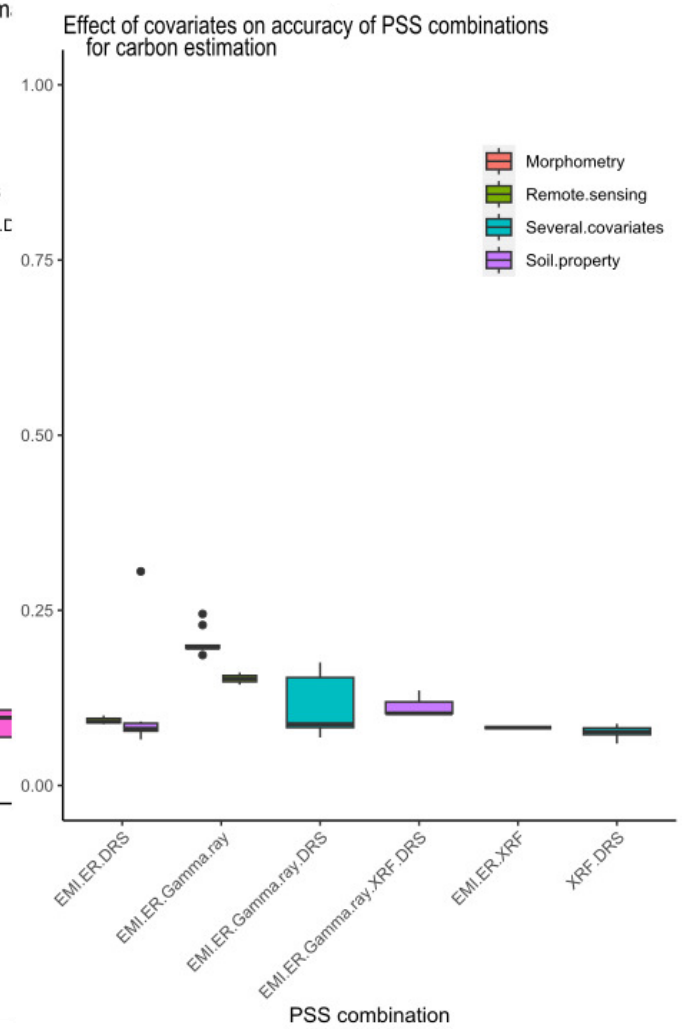
N records = 321



N records = 96
N records $ soil property = 47
N references = 10



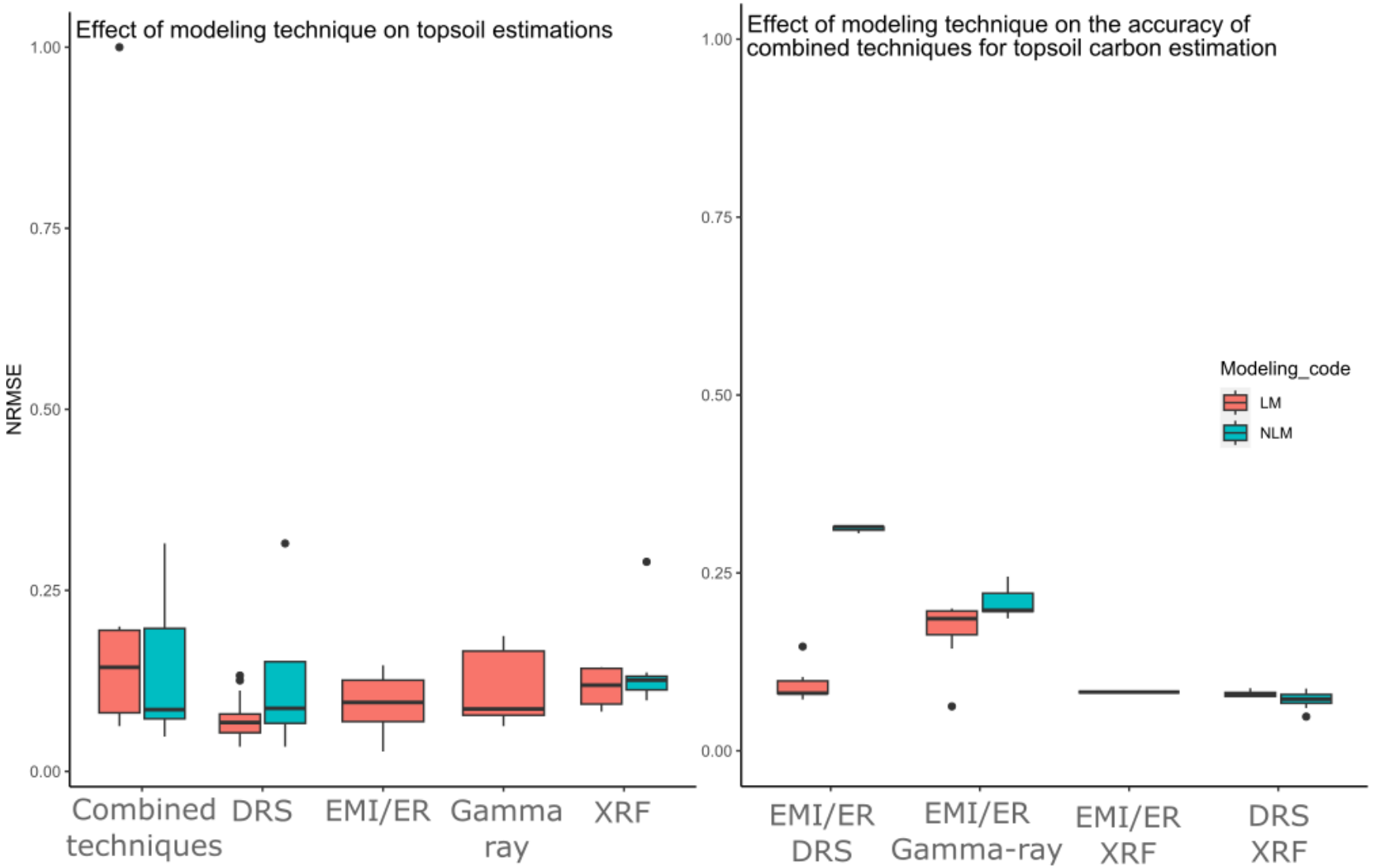
N records = 75



N records = 46

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TOPSOIL SOC ESTIMATION



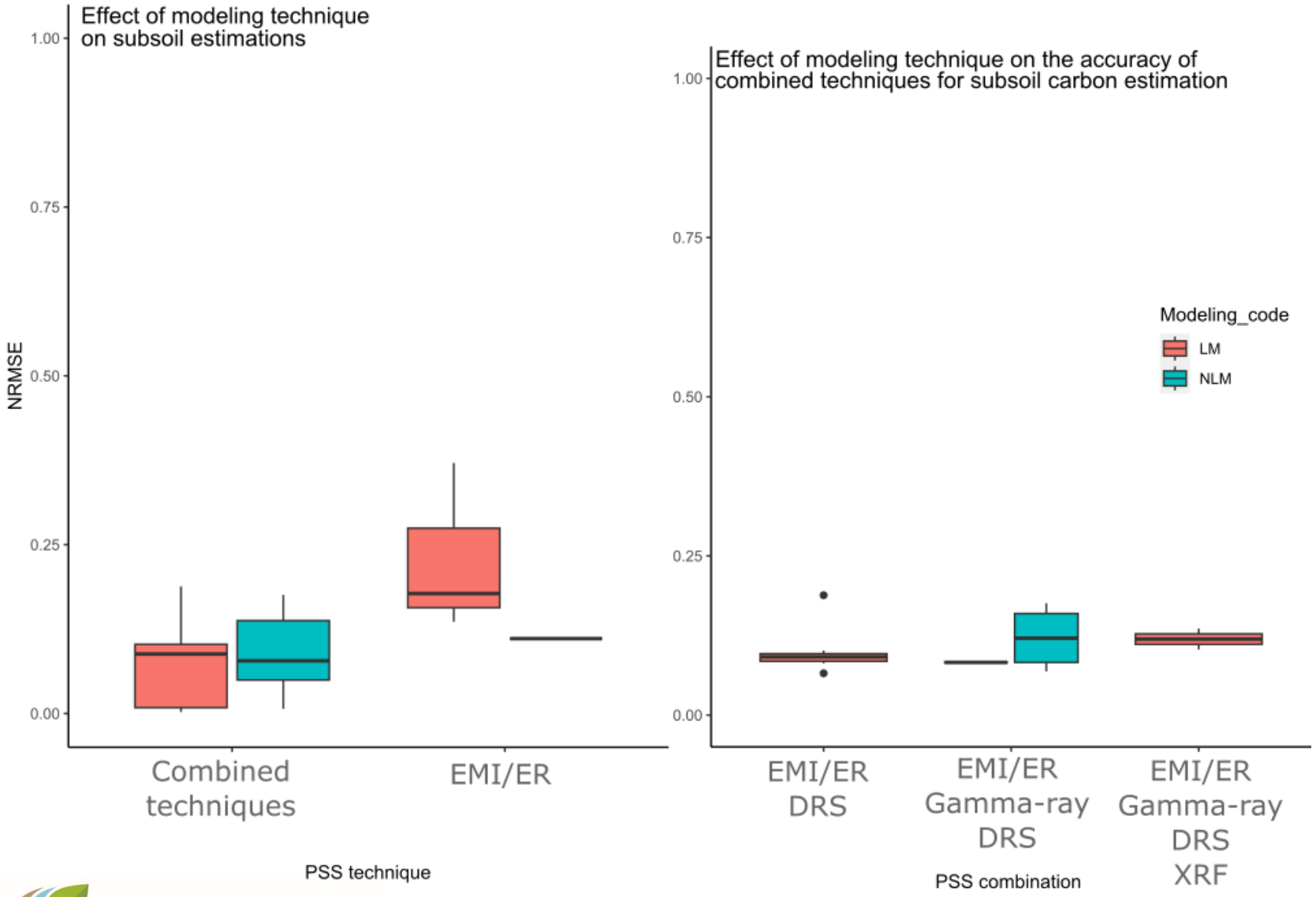
An improvement in accuracy was not found when non-linear modeling techniques were applied

When several sensors are combined through data fusion, the performance of NLM is lower

LM: PLSR, MLR, etc.
NLM: RT, SVM, RF, etc.

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SUBSOIL SOC ESTIMATION



An improvement in accuracy was not found when non-linear modeling techniques were applied

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LM: PLSR, MLR, etc.
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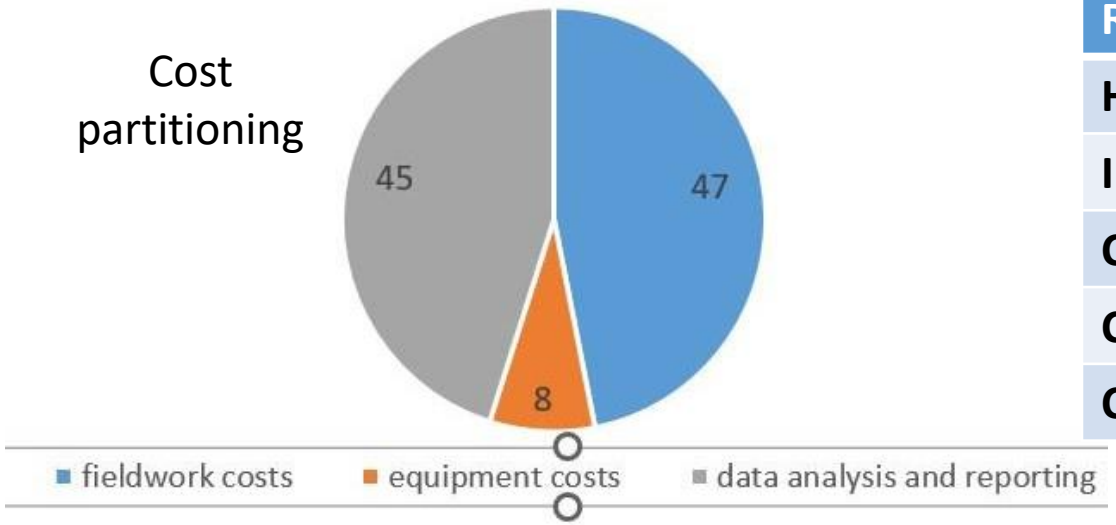
Best combo for each group of properties

Group of soil properties	Best Combo			Number of papers	NRMSE			
	PSS technique	Model technique	Covariates		Min	Mean	SD	Max
Acidic-basic properties	DRS	Linear	Yes	16	0.038	0.081	0.025	0.117
Bulk density	Combined techniques	Linear	Yes	7	0.069	0.095	0.027	0.139
Carbon	Combined techniques	Linear	No	17	0.002	0.081	0.064	0.188
Hydrological properties	Combined techniques	Non-linear	No	8	0.025	0.052	0.031	0.108
Mineralogy	XRF	Non-linear	No	18	0.009	0.501	0.474	1.067
Nitrogen	XRF	Linear	Yes	7	0.074	0.134	0.062	0.250
Nutrients+CEC+ex.bases	DRS	Non-linear	No	11	0.035	0.089	0.047	0.178
Profile features+other prop.	DRS	Linear	No	10	0.017	0.105	0.039	0.160
Soil depth	EMI/ER	Linear	No	12	0.099	0.121	0.023	0.168
Texture	XRF	Linear	No	5	0.074	0.093	0.018	0.113

PRELIMINARY RESULTS: COSTS ANALYSIS

11 Countries: Austria, Canada, Denmark, France, Italy, Netherlands, Spain, Sweden, Switzerland, Turkey and USA
24 companies invested in multiple types of sensors
Texture mapping was the most requested service
The most frequent technique was **EMI followed by DRS** (on-the-go, VERIS tech.)

Cost partitioning



Report	Min	Q1	Median	Q3	Max
Ha/day	1	15	30	100	300
Interline space	5	13	20	125	250
Cost/Ha	100	120	142	362	1,300
Cost/day	400	2,575	4,510	7,025	10,070
Cost renting	60	300	300	600	800

Companies declared
No extra charged costs depending on the technique used
No extra charged costs due to difficulties (e.g. accessibility, field conditions)

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TAKE-HOME MESSAGES

From literature search:

- Almost half of the predictive soil modelling studies are based on DRS.
- While DRS showed superior performance compared to other techniques, it is important to acknowledge that most of the studies are laboratory-based. In addition, those predictions were property estimations in soil samples, not soil maps.
- Non-linear modelling does not offer a significant advantage compared to linear modelling.
- Sensor fusion can at times be advantageous.
- The role of other covariates in improving the predictability is case-specific.

From company surveys:

- On-to-go sensors (EMI, Gamma and shank-based NIR) are more popular.
- Texture is the most requested soil property and 4 samples per ha are commonly collected for calibration.
- The cost of PSS service is around 362 Eur/Ha on average and a significant proportion of the costs are for fieldwork and data analysis with minimal costs for reporting.

SPOILER ALERT!!

When completed, the analysis of data will throw the best selection of methods depending on the circumstances to collect soil information with the right accuracy at the lowest cost.

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INIA Instituto Nacional de Investigación y Tecnología Agraria y Alimentaria

Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

ptj Projektträger Jülich Forschungszentrum Jülich

INRAE

BIOS science austria

WAGENINGEN UNIVERSITY & RESEARCH
ILVO Institute for Agricultural and Fisheries Research

ATK Talajtani és Agrokémiai Intézet

eagasc Agriculture and Food Development Authority

crea Consiglio per la ricerca in agricoltura e l'analisi dell'economia agraria

Wallonie recherche
CRA-W

AARHUS UNIVERSITY
DCA - DANISH CENTRE FOR FOOD AND AGRICULTURE

SLU

afbi AGRI-FOOD & BIOSCIENCES INSTITUTE

University of Latvia

LITHUANIAN RESEARCH CENTRE FOR AGRICULTURE AND FORESTRY

Czech University of Life Sciences Prague

Eesti Maaülikool
EMU Estonian University of Life Sciences

TAGEM R&D AND INNOVATION

NIBIO Norsk institutt for bioøkonomi

Instituto Nacional de Investigação Agrária e Veterinária, I.P.

IUNG

NPPC National Agricultural and Food Centre

Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Swiss Confederation

Federal Department of Economic Affairs, Education and Research EAER

Agroscope

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life from cosmos
finapp

GEOPROSPECTORS

Veris technologies
+ SOIL MASTERS

SOING
STRUTTURE & AMBIENTE

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To all companies, and those who preferred remaining anonymous

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Thank you for your time and attention

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